Zomato Bangalore Restaurant Data Science Project

Final Internship Report

Data analysis, machine learning and business recommendations

Prepared by	Program	Project scope
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Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn and Jupyter Notebook

Executive Summary

The analysis converts a noisy 56,252-row Zomato export into a decision-ready dataset of 33,660 valid listings. The strongest market signal is a large value-oriented delivery segment, while a smaller premium dine-in segment achieves higher ratings, engagement, prices and booking adoption. The machine-learning results are strong enough for portfolio demonstration, but restaurant-level deployment requires grouped validation to prevent repeated listings of the same restaurant from appearing across training and test data.

Measure	Result	Interpretation
Clean listings	33,660	40.2% fewer than raw after structural validation and deduplication
Average rating	3.68 out of 5	Most valid ratings sit in the mid-to-high range
Median cost for two	INR 400	Market is strongly value and mid-range oriented
Online ordering	60.21%	Digital ordering is already mainstream
Table booking	11.58%	Reservation capability is concentrated in dine-in formats
Best classifier	Random Forest, F1 0.923	Predicts above-median price class; ROC AUC 0.972
Best regressor	Extra Trees, RMSE 0.161	Explains 85.8% of rating variance on the notebook split

Priority Recommendations

- Build separate strategies for the premium dine-in and value delivery segments instead of using one citywide operating model.
- Treat BTM, Whitefield, HSR, Marathahalli and Indiranagar as high-activity but high-competition markets; validate demand gaps before expansion.
- For value restaurants, improve digital menu accuracy, packaging, preparation time and delivery reliability. For premium restaurants, strengthen reservations, service consistency and review management.
- Before production use, consolidate restaurant identities, use GroupKFold or a group train-test split, and test performance on a later time period or an external dataset.

Decision takeaway: the project provides a credible market and modeling baseline, with the clearest immediate value coming from segment-specific operating decisions rather than automated prediction alone.

Project Context and Objectives

This internship project examines restaurant supply, service options, price, ratings, votes, cuisines and location concentration in Bangalore. The objective is to convert a real-world restaurant export into reliable descriptive insights, predictive models and actionable recommendations for restaurant owners and food-platform teams.

Objective	Analytical response
Understand the market	KPIs, distributions, top locations, restaurant formats and cuisines
Explain service and rating patterns	Grouped comparisons, box plots and correlation analysis
Predict price segment	Binary classification of cost above the median
Predict restaurant rating	Regression using service, engagement, cost and category features
Create strategic segments	K-Means clustering with PCA and t-SNE visualization

Notebook and Submission Files

Primary notebook: Alfido_Tech_Zomato_Complete_Data_Science_Project.ipynb

Dataset: zomato(3).csv. The notebook contains 145 cells covering data validation, preprocessing, EDA, six classification models, eight regression models, hyperparameter tuning, feature importance, K-Means, DBSCAN, PCA and t-SNE.

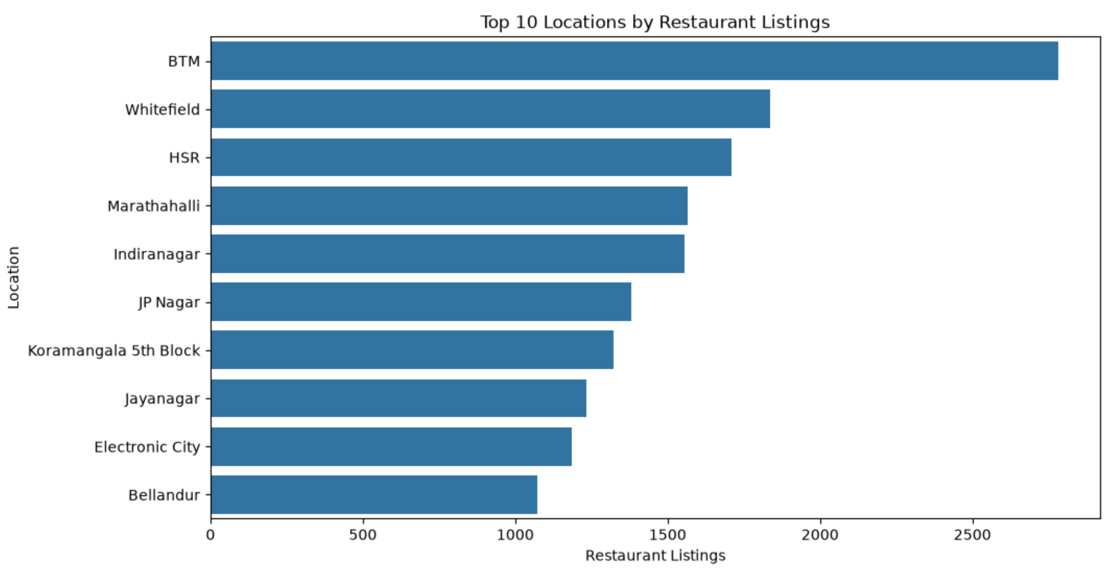

Figure 1. Executed notebook output for the top location analysis.

Dataset and Data Quality

Field group	Included variables	Purpose
Identity and geography	name, address, phone, location	Restaurant reference and market concentration
Service	online_order, book_table	Digital ordering and reservation availability
Performance	rate, votes	Customer rating and engagement
Offer and format	rest_type, cuisines, dish_liked, listed_in(type)	Restaurant positioning and menu representation
Price	approx_cost(for two people)	Affordability and price-segment modeling

Cleaning Decisions

- Structural validation retained 47,356 rows whose service and listing-category values matched known domains; 8,896 malformed rows were removed.
- Exact deduplication removed 13,696 additional rows from the structurally valid subset, leaving 33,660 analytical listings.
- Ratings were extracted from text, converted to numeric values and constrained to the 0-5 range. This yielded 28,133 valid rating targets.
- Votes and cost were converted to numeric values; commas were removed from cost. Missing predictors were handled within model pipelines.
- dish_liked remained 53.47% missing after cleaning and was excluded from the primary predictive feature sets.

Quality Assessment

The raw file contained 15,703 exact duplicate rows (27.92%). The difference between raw duplicates and duplicates removed after structural filtering is expected because malformed rows are removed first. The cleaned file represents listings rather than guaranteed unique physical restaurants; 8,641 distinct restaurant names appear across 33,660 rows.

Implication: descriptive counts should be described as listings, and predictive validation should account for repeated restaurant identities.

Market Analysis and Key Findings

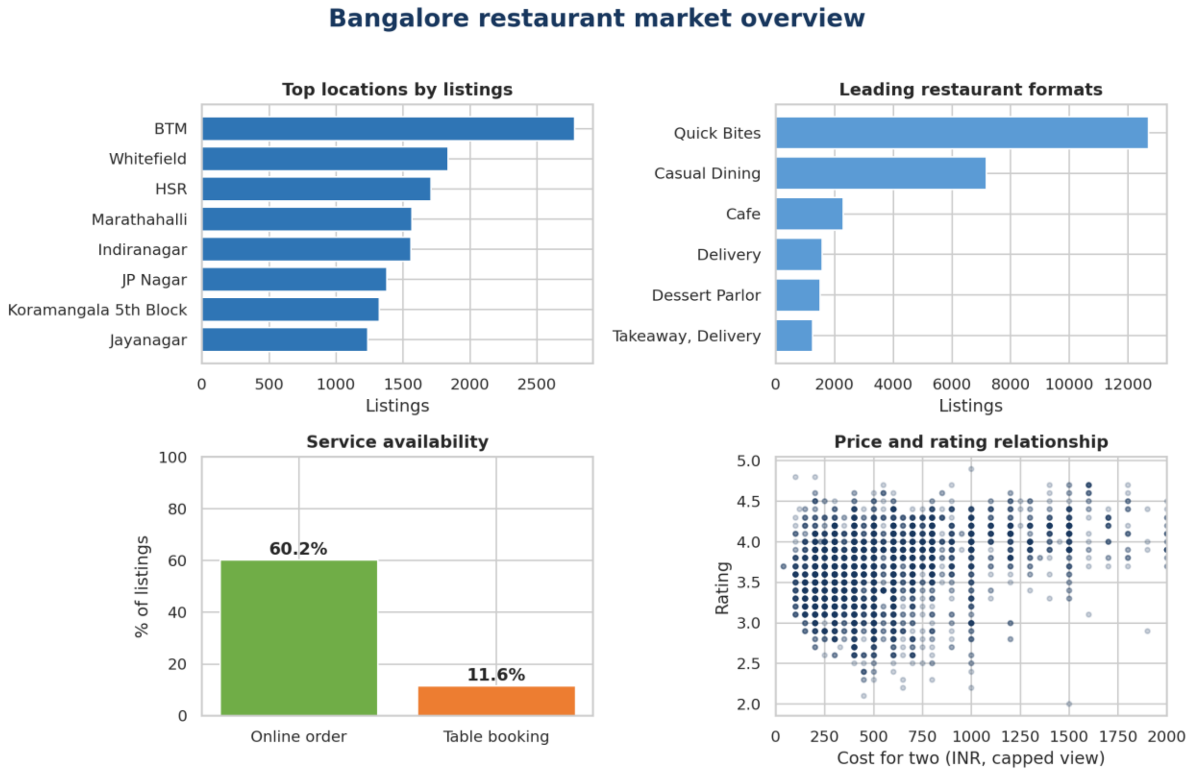

Figure 2. Market concentration, restaurant formats, service availability and the price-rating relationship.

Finding	Evidence	Business meaning
BTM leads location supply	2,781 listings	Strong activity and competition; opportunity requires gap analysis
Quick Bites dominates	12,691 listings	Convenience and value formats are central to the market
North Indian leads cuisines	13,912 appearances	Large addressable demand, but likely intense competitive density
Delivery is the top listing category	16,221 listings	Off-premise demand is structurally important
Online order correlates with slightly higher ratings	3.705 vs 3.629	Association is small and may reflect restaurant mix
Table booking correlates with higher ratings	4.078 vs 3.617	Likely reflects premium format and price confounding

Cost and rating correlate at 0.334, while votes and rating correlate at 0.399. These moderate relationships support their use as predictors, but neither proves that raising prices or generating more votes will cause higher satisfaction.

Predictive Modeling

All models use preprocessing pipelines with categorical imputation and one-hot encoding plus numerical median imputation. The classification split is stratified; both tasks use an 80:20 train-test split with random_state 42. Five-fold cross-validation is used as a reliability check.

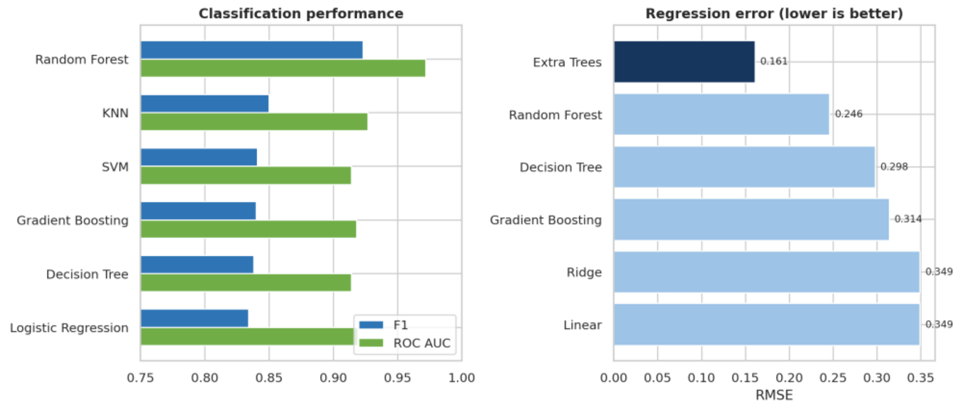

Figure 3. Classification and regression comparison from executed notebook results.

Classification of Above Median Cost

Model	Accuracy	Precision	Recall	F1	ROC AUC
Random Forest	0.928	0.950	0.897	0.923	0.972
KNN	0.862	0.885	0.817	0.850	0.927
SVM	0.858	0.906	0.784	0.841	0.914

Random Forest leads with five-fold F1 of 0.9181 ± 0.0024. The target threshold is the cleaned median cost of INR 400; cost_for_two is excluded to prevent direct leakage.

Regression of Restaurant Rating

Model	MAE	RMSE	R2
Extra Trees	0.065	0.161	0.858
Random Forest	0.164	0.246	0.669
Decision Tree	0.204	0.298	0.515

Extra Trees leads; cross-validated RMSE is 0.1515 ± 0.0052. Treat this as a portfolio benchmark until group-aware and time-aware validation confirms performance on unseen restaurants.

Feature Importance and Segmentation

Most Useful Predictive Signals

For the tuned Random Forest classifier, votes (0.182), Casual Dining (0.126), Quick Bites (0.116), rating (0.104) and cuisine count (0.065) are the leading features. For rating prediction, votes has the largest reported importance (0.533), followed by cost for two (0.085) and cuisine count (0.040). Importance describes model usage, not causality.

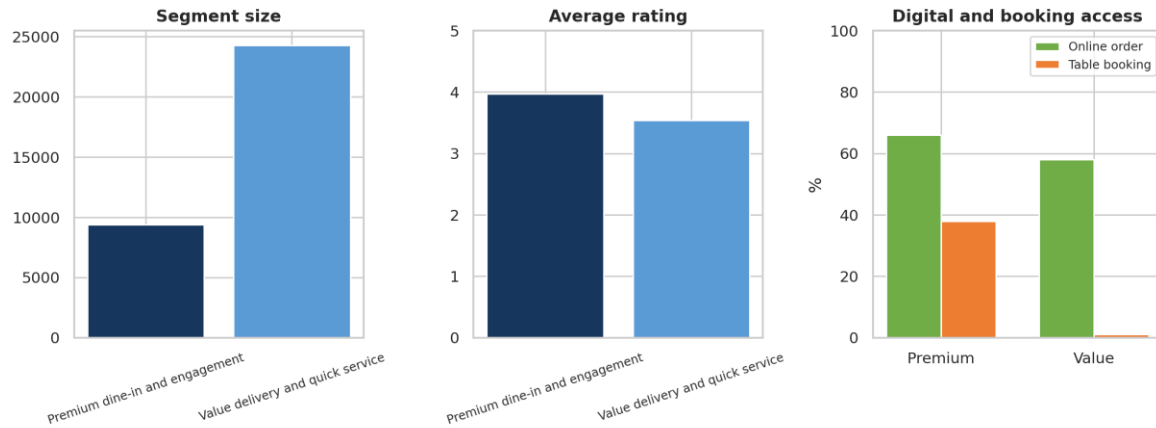

Figure 4. Two-cluster profile selected by the highest sampled silhouette score of 0.346.

Profile	Premium dine-in and engagement	Value delivery and quick service
Listings	9,378	24,282
Average rating	3.97	3.54
Average votes	590.69	80.46
Average cost for two	INR 957.74	INR 385.51
Online order rate	66%	58%
Table booking rate	38%	1%
Typical format	Casual Dining, Dine-out	Quick Bites, Delivery
Typical location	Koramangala 5th Block	BTM

The two segments are strategically distinct. The premium segment combines higher price, engagement, ratings and reservation usage. The value segment is much larger and optimized around quick service and delivery. The segment names are business interpretations of statistical clusters and should be validated with operators.

Business Recommendations

1 Location portfolio

Score candidate micro-markets on listing density, cuisine saturation, price bands, ratings and competitor engagement. High density is evidence of demand and competition, not a stand-alone expansion signal.

2 Value delivery operations

Prioritize menu clarity, packaging, prep-time control, availability accuracy and service recovery. Track cancellation rate, delivery time, repeat orders and rating movement by location.

3 Premium dine-in experience

Use reservation tools, peak-hour capacity planning, differentiated menus and consistent front-of-house service. Track seat utilization, booking conversion, no-shows, average bill value and review themes.

4 Cuisine differentiation

Treat North Indian and Chinese as large but crowded categories. Test narrower concepts, signature dishes, bundles and location-specific menu variants before broad rollout.

5 Responsible model use

Use the classifier for market research and prioritization, not automatic pricing. Use rating predictions to identify cases for review rather than presenting them as guaranteed outcomes.

6 Measurement plan

Run controlled pilots by location and segment. Compare pre/post and matched-control outcomes for ratings, order volume, contribution margin, repeat rate and complaints.

Recommended 90 Day Action Plan

Period	Action	Success measure
Days 1-30	Consolidate restaurant identities; build location-cuisine-price opportunity matrix	Duplicate identity rate and complete baseline KPIs
Days 31-60	Pilot one value-delivery and one premium-dine-in intervention	Operational KPIs improve without margin deterioration
Days 61-90	Revalidate models with grouped splits and compare pilot cohorts	Stable out-of-group metrics and measurable business lift

Limitations and Validation Requirements

Risk	Why it matters	Required control
Unit of analysis	Rows are listings, not confirmed unique establishments. The same restaurant can appear in multiple categories or locations.	Create a stable restaurant ID
Potential split leakage	Random row splitting can place related or repeated listings in both train and test sets, inflating model performance.	Use group-aware train-test split
Observational data	Associations between services, cost, votes and ratings do not establish cause and effect.	Use experiments or causal designs
Missingness	Ratings and liked dishes have substantial missingness; missingness may not be random.	Analyze missingness and sensitivity
Time and geography	The dataset lacks a clearly modeled observation date and is specific to Bangalore; market conditions can change.	Validate on recent external data
Target design	The expensive target is relative to this dataset median, so INR 400 is not a universal business threshold.	Recalibrate threshold by market and objective
Cluster quality	A silhouette score of 0.346 indicates useful but moderate separation; segments overlap and need business validation.	Profile and test segment stability

Reproducibility

1. Place the notebook and CSV in the same folder.
2. Rename the dataset to zomato.csv or update DATA_PATH in the loading cell.
3. Create a Python environment and install pandas, numpy, matplotlib, seaborn, scikit-learn and jupyter.
4. Run all notebook cells from top to bottom and confirm model metrics before submission.
5. Export the cleaned analytical dataset only after validation and keep the raw file unchanged.

Conclusion

This project demonstrates an end-to-end data science workflow: business framing, data validation, cleaning, KPI design, exploratory analysis, feature engineering, supervised learning, tuning, feature importance, unsupervised segmentation and business interpretation. The strongest commercial conclusion is that Bangalore restaurant listings separate into a large value-delivery market and a smaller, higher-engagement premium dine-in market that require different operating strategies.

The executed notebook shows strong predictive performance, especially from Random Forest classification and Extra Trees regression. The next step is not additional algorithm complexity; it is stronger validation at the restaurant identity and time levels. With that improvement, the project becomes a more credible decision-support prototype as well as a strong internship portfolio submission.

Appendix Project Metrics

Category	Metric	Value
Data	Raw rows	56,252
Data	Clean listings	33,660
Data	Unique restaurant names	8,641
Data	Locations	93
Market	Top location	BTM, 2,781 listings
Market	Top type	Quick Bites, 12,691 listings
Market	Top cuisine	North Indian, 13,912 appearances
Classification	Random Forest	F1 0.923, ROC AUC 0.972
Regression	Extra Trees	RMSE 0.161, R2 0.858
Clustering	K-Means	k = 2, silhouette 0.346

Source note: all statistics and model results in this report are taken from the supplied CSV and executed notebook outputs. Currency values represent the dataset field for approximate cost for two people.